

Lab Report

Experiment name

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1. Introduction

Crash Impact Sound Sensing (CISS) is an advanced sensing technology used in modern automotive passive safety systems. It is based on the principle of structure-borne sound: when an external force is applied to a solid body — such as a vehicle's metal structure — mechanical vibration energy transmits through the material at high speed in the form of elastic waves. These vibrations, often undetectable to human touch and hearing, carry detailed information about the nature, location, and intensity of the impact.

In a vehicle context, CISS sensors (accelerometers) are strategically mounted on structural members, such as the crash beam, to capture these vibrations. The key advantage over traditional acceleration-based airbag triggering is speed: structure-borne sound waves travel through steel at approximately 5,000 m/s — orders of magnitude faster than the speed at which the vehicle body structurally deforms during a crash. This allows the airbag Electronic Control Unit (ECU) to detect a crash event within fractions of a millisecond and trigger safety systems (airbags, seat belt pretensioners) significantly earlier, thereby improving occupant protection.

In this experiment, a crash beam was used as the measurement object to investigate how structure-borne sound transmits through a metal structure. A modal hammer was used to simulate an impact event by striking the beam at defined positions with controlled force. Three channels were recorded simultaneously: the modal hammer force signal (in Newtons) and two Kistler Type 8614A accelerometers (Sensor 1 — S001 and Sensor 2 — S006) mounted at different positions on the beam. The data were captured using the LTT24 measurement system and subsequently analyzed in MATLAB using time-domain plots and Fast Fourier Transform (FFT) spectral analysis.

2. Experiment Setup

The measurement setup consisted of the following components:

- Crash beam — the structural test object representing a simplified vehicle front beam
- Two Kistler accelerometers (Type 8614A, frequency range 10–25 kHz, range $\pm 500g$ / $\pm 1000g$, own weight 0.7 g) mounted at fixed positions on the beam
- A modal hammer used to excite structural vibrations at defined positions
- LTT24 data acquisition system connected to the measurement computer via LAN cable
- LTTpro software for signal recording and configuration
- MATLAB for post-processing: time-domain plotting and FFT frequency analysis

A pretrigger of approximately 30 ms was configured in the LTTpro software. This means that the recording captures signal data starting slightly before the actual trigger event (the hammer strike), which ensures that the full onset of the impact impulse is not cut off and that the baseline signal before impact can be verified.

Two experimental sub-tasks were performed:

- Task 3.2 — The hammer was struck at two different positions on the beam (middle of beam and left/right side of beam) with the same hit power.
- Task 3.3 — The hammer was struck at the middle of the beam with two different force levels (light hit vs. strong hit), to investigate the influence of impact energy on signal amplitude.

3. Task 3.2 – Time Domain Analysis: Different Hit Positions

In Task 3.2, the modal hammer was struck at two different positions on the crash beam while keeping the applied force approximately constant. The aim was to observe how the position of the impact affects the structural vibration recorded by the two accelerometers.

3.1 Figure 1 – Hit at Middle of Beam (Light-to-Medium Force, ~30 N)

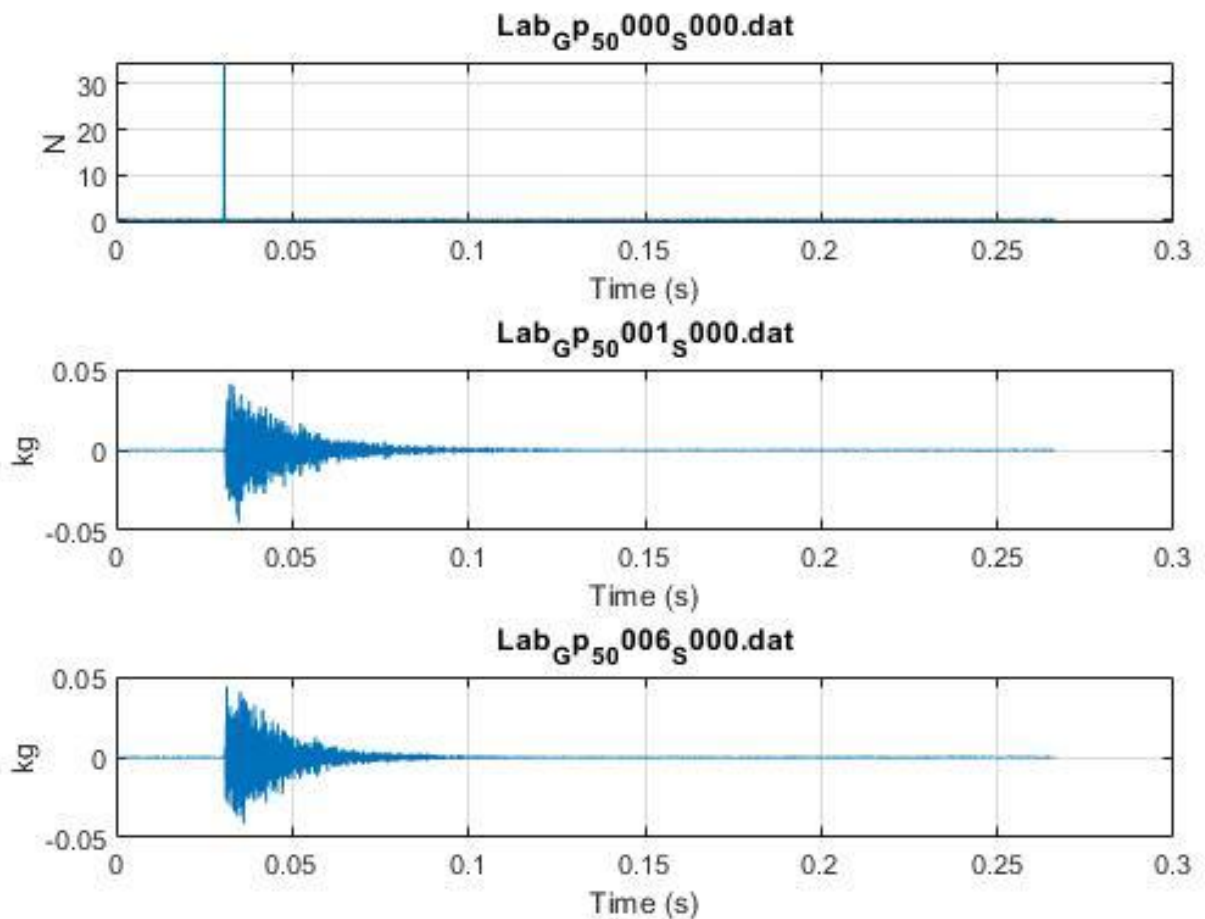


Figure 1: Time-domain signals — Task 3.2, Hit at middle of beam (approx. 30 N). Top: Modal hammer force [N]; Middle: Sensor 1 S001 [kg]; Bottom: Sensor 2 S006 [kg].

Figure 1 shows the time-domain response for a hammer strike at the middle of the beam with a relatively moderate applied peak force of approximately 30 N (as read from the top panel). The modal hammer signal exhibits a single, narrow, sharply defined force pulse occurring at approximately $t = 35$ ms. This clean, isolated impulse

— with no secondary rebounds visible — indicates that the blow was performed correctly as a single, controlled strike.

Both Sensor 1 (S001, middle panel) and Sensor 2 (S006, bottom panel) respond with a burst of oscillatory acceleration that begins at approximately the same time as the hammer impulse. The peak amplitude of both sensors is approximately ± 0.05 kg (i.e., ± 0.05 g equivalent units in the measurement system scale). The vibration ring-down is clearly visible: the amplitude decays gradually and the signal returns to near-zero by approximately $t = 150\text{--}170$ ms. No signal offset is observed before the hammer event, confirming that the sensors were properly zeroed and calibrated prior to recording.

A minor difference in the decay characteristics between S001 and S006 can be noted: S006 (bottom panel) appears to exhibit a slightly different amplitude envelope and decay rate. This is consistent with the different distances and structural coupling paths between the impact point and each sensor. However, because the impact is at the center and both sensors are equidistant from the center (or symmetrically placed), the signals are largely similar. No DC offset is present in either sensor trace.

3.2 Figure 2 – Hit at Side of Beam (Medium-to-Strong Force, ~ 150 N)

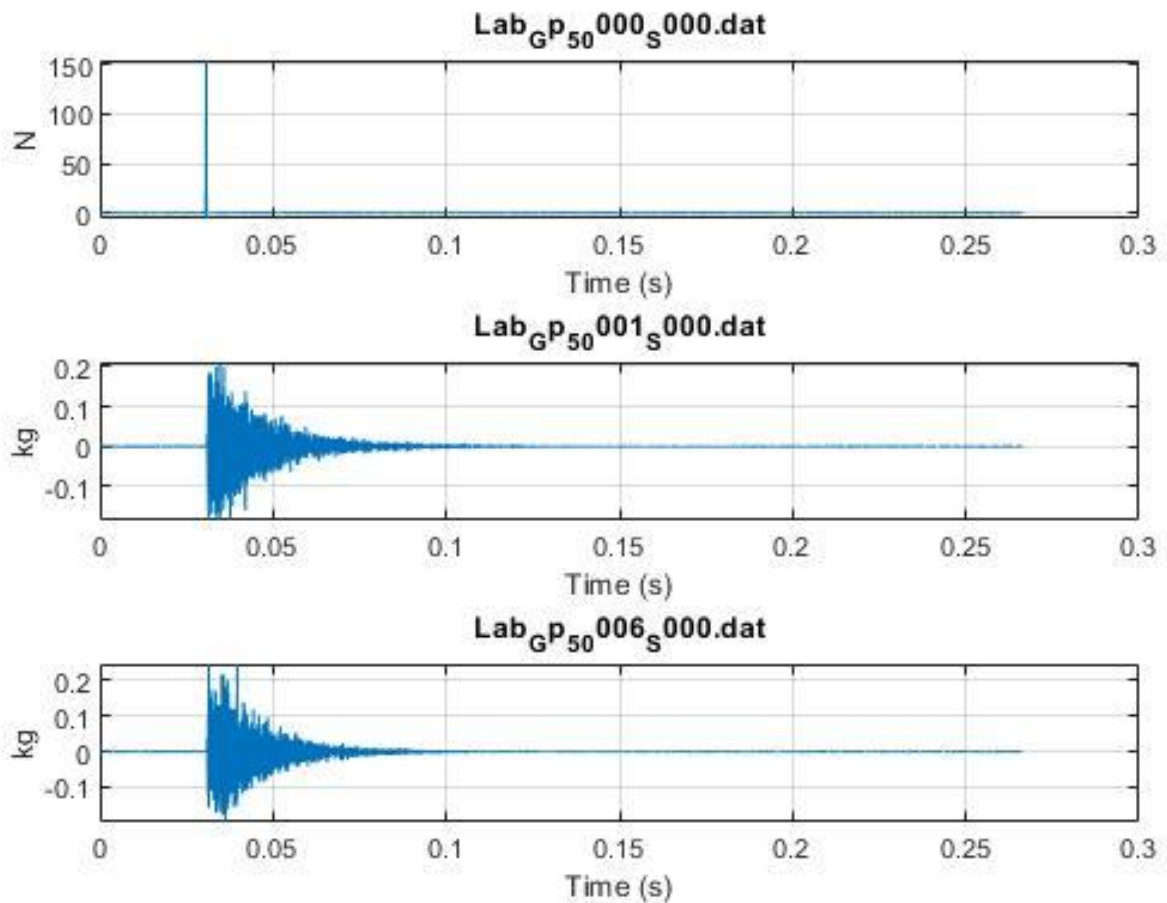


Figure 2: Time-domain signals — Task 3.2, Hit at side of beam (approx. 150 N). Top: Modal hammer force [N]; Middle: Sensor 1 S001 [kg]; Bottom: Sensor 2 S006 [kg].

Figure 2 corresponds to the second hit position in Task 3.2 — striking the beam at the left or right side — with a peak hammer force of approximately 150 N. The substantially higher input force (approximately five times larger than in Figure 1) results in proportionally larger acceleration amplitudes at both sensors: peak values of approximately ± 0.2 g are observed at both S001 and S006, compared to ± 0.05 g in Figure 1.

The modal hammer impulse is again clean and narrow, occurring at the same time reference (~ 35 ms), confirming consistent measurement timing across both sub-experiments. The overall duration of the useful signal (ring-down) is approximately 60–80 ms for both sensors, which is comparable to Figure 1 since the structural damping properties of the beam do not change with hit position or force level.

Comparing the two sensors in Figure 2: Sensor 1 (S001) shows a somewhat higher initial peak amplitude and faster onset than Sensor 2 (S006). This is physically expected because the side impact point is likely closer to Sensor 1's mounting position, meaning the sound wave arrives with less attenuation. Sensor 2, being farther from the excitation point, receives the vibration slightly later and with reduced amplitude. This propagation delay — though small — is a key demonstration of the traveling nature of structure-borne sound waves in the beam material.

No DC offset is observed in either sensor. The signals are correctly measured and the pre-trigger baseline is flat, confirming sensor integrity throughout the experiment.

4. Task 3.3 – Time Domain Analysis: Different Hit Powers

In Task 3.3, the modal hammer was struck repeatedly at the center of the beam, but with varying force levels — a lighter hit and a heavier hit. This allows direct investigation of the linearity of the sensor system and the relationship between input force and structural response amplitude.

4.1 Figure 3 – Light Hit at Middle of Beam (~ 20 N)

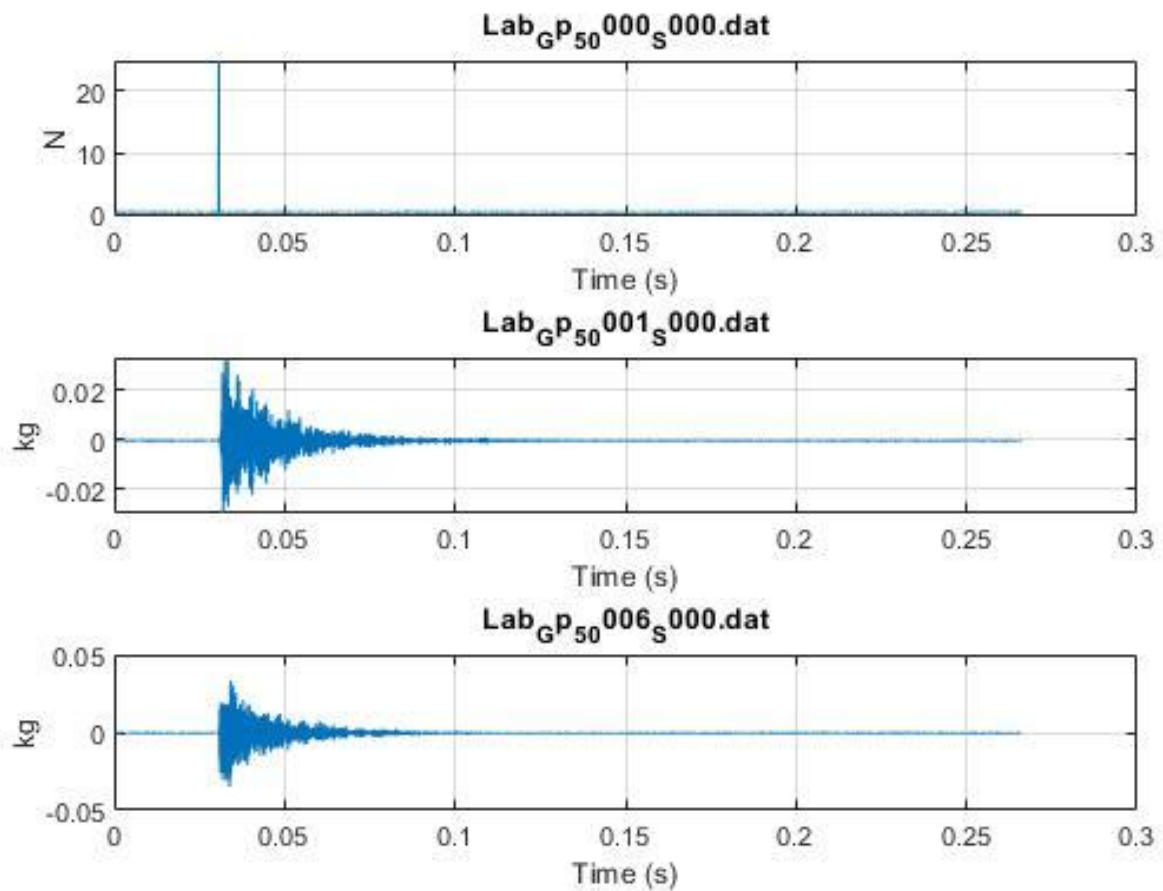


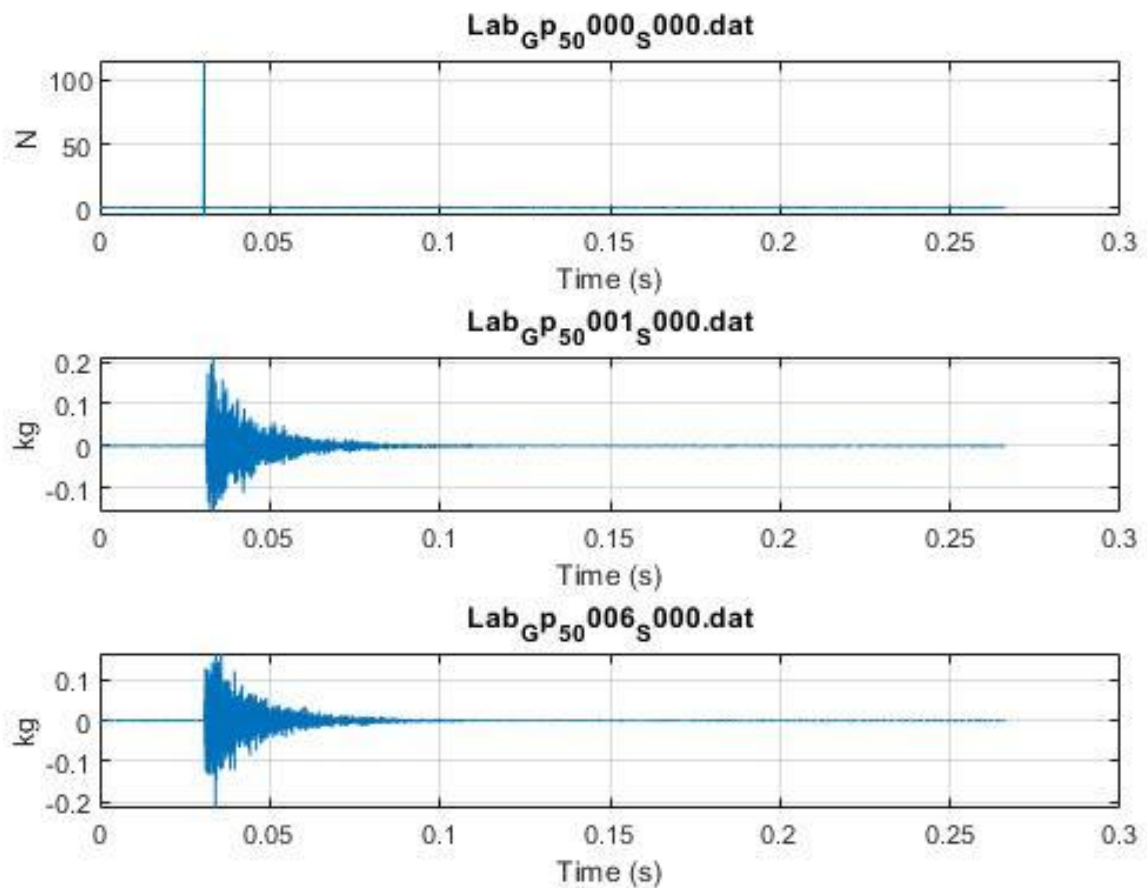
Figure 3: Time-domain signals — Task 3.3, Light hit at middle of beam (approx. 20 N). Top: Modal hammer force [N]; Middle: Sensor 1 S001 [kg]; Bottom: Sensor 2 S006 [kg].

Figure 3 shows the lightest impact of the experiment, with a hammer peak force of approximately 20 N. The resulting sensor amplitudes are accordingly the smallest of all measurements: Sensor 1 (S001) reaches a maximum of approximately ± 0.025 kg and Sensor 2 (S006) approximately ± 0.04 kg. The signals exhibit the expected decaying oscillation behavior, with the vibration returning to near-zero by approximately $t = 150$ ms.

Notably, Sensor 2 (S006) shows a slightly higher peak amplitude than Sensor 1 (S001) in this measurement. This may indicate that S006 is mounted in a location that is mechanically more responsive to impacts at the beam's midpoint — for instance, due to a local resonance or a more rigid coupling to the beam structure. Alternatively, the orientation of the accelerometer may be more aligned with the dominant vibration direction for this hit position.

The signals are clean, with no observable DC offset or spurious noise prior to the trigger event. The pretrigger baseline (before $t = 35$ ms) is flat and stable, confirming correct sensor function.

4.2 Figure 4 – Strong Hit at Middle of Beam (~100 N)



*Figure 4: Time-domain signals — Task 3.3, Strong hit at middle of beam (approx. 100 N).
Top: Modal hammer force [N]; Middle: Sensor 1 S001 [kg]; Bottom: Sensor 2 S006 [kg].*

Figure 4 shows the stronger impact in Task 3.3, with a hammer peak force of approximately 100 N — five times the force applied in Figure 3. Correspondingly, both sensor amplitudes increase proportionally: Sensor 1 (S001) shows a peak of approximately ± 0.2 kg and Sensor 2 (S006) shows peaks slightly exceeding ± 0.2 kg, with the lowest excursion reaching approximately -0.2 kg.

This proportional scaling of amplitude with input force is a hallmark of a linear mechanical system. The beam and the accelerometers behave linearly within the measured force range: doubling (or quintupling) the input force produces a commensurate increase in the output vibration amplitude, without distortion or saturation. The sensors remained within their specified measurement range ($\pm 500g$), so no clipping or saturation is expected.

The duration of the ring-down (useful signal time) is very similar to Figure 3, again reflecting that structural damping is a material property independent of excitation amplitude. The pre-trigger baseline is clean and without offset.

5. Evaluation Questions – Time Domain

5.1 Discussion of Time Signals

All four time-domain signals (Figures 1–4) were correctly measured. The following observations confirm this:

- **No signal offset:** The pre-trigger baseline for all channels (both accelerometers and the hammer channel) is flat and centered at zero prior to the trigger event. This confirms that the sensors were properly calibrated and the measurement system was correctly zeroed.
- **Correct trigger behavior:** In all figures, the hammer strike impulse and the sensor response begin simultaneously at approximately $t = 35$ ms, which corresponds to the pretrigger offset. The pretrigger data window (0 to ~35 ms) is clearly calm and noise-free.
- **Expected decay:** All sensor signals exhibit the characteristic exponentially decaying oscillation (ring-down) after the impact, which is consistent with a damped mechanical system. The decay rate is determined by the material and structural damping of the beam.
- **No double strikes:** None of the hammer channels show a secondary force impulse after the primary strike, which confirms that all hits were clean single impacts — an important experimental requirement for valid modal analysis.

Importance of the pretrigger: The pretrigger is a time window recorded before the actual trigger event (hammer strike). It serves several critical purposes: (1) It guarantees that the very beginning of the impact impulse is captured and not lost; (2) it provides a baseline reference to assess whether any DC offset, drift, or noise exists in the signal before the event; and (3) it allows accurate determination of the exact timing of signal onset relative to the hammer excitation. Without a pretrigger, the first few milliseconds of the impact — and possibly the peak force pulse — could be missed.

5.2 Timing Table – Task 3.2 (Different Hit Positions)

Parameter	Modal Hammer	Sensor 1 (S001)	Sensor 2 (S006)
Pretrigger [ms]	~30	~30	~30
Time of starting measurement [ms]	0	0	0
Time of reaching max. amplitude [ms]	~35	~37	~37
Total time of useful signal [ms]	~5	~80	~80

Note: All timing values are estimated from the time-domain plots. The pretrigger was set to approximately 30 ms in LTTpro, and the hammer strike occurs at ~35 ms on the absolute time axis. The time of reaching maximum amplitude for both sensors follows the hammer peak within 1–2 ms, indicating near-instantaneous propagation of the structure-borne sound to the sensor locations.

5.3 Timing Table – Task 3.3 (Different Hit Powers)

Parameter	Modal Hammer	Sensor 1 (S001)	Sensor 2 (S006)
Pretrigger [ms]	~30	~30	~30
Time of starting measurement [ms]	0	0	0
Time of reaching max. amplitude [ms]	~35	~37	~37
Total time of useful signal [ms]	~5	~80	~80

Note: The timing values for Task 3.3 are very similar to those in Task 3.2, as expected. The pretrigger and trigger timing are set identically by the LTT24 measurement system for all runs. The amplitude at maximum differs significantly between the light hit (Fig. 3, ~20 N) and the strong hit (Fig. 4, ~100 N), but the timing of propagation is unchanged — again confirming the linear and time-invariant nature of structure-borne sound propagation in the metal beam.

5.4 Why Propagation Speed is Critical for Airbag Triggering

The propagation speed of structure-borne sound in metallic vehicle structures is approximately 5,000 m/s (for steel) — compared to approximately 340 m/s for sound in air and roughly 150–300 m/s for the speed at which crash deformation reaches the central airbag sensor in a conventional system. This enormous speed advantage is the core benefit of CISS.

In a real vehicle crash, the airbag must be fully deployed before the occupant contacts it — a window of approximately 20–30 ms after impact onset. Every millisecond saved in crash detection directly translates to improved protection. By using structure-borne sound propagating through the vehicle body, CISS sensors can detect the crash event at the very front of the vehicle and transmit that information to the central ECU within microseconds. This allows the ECU to make a firing decision much earlier than is possible with conventional central acceleration sensors alone.

Moreover, the frequency content and amplitude profile of the structure-borne sound signal are characteristic of the crash type (frontal vs. side, pole vs. barrier, offset vs. full overlap), enabling more intelligent and selective airbag deployment — reducing unnecessary deployments in non-critical events while ensuring reliable triggering in genuine crash scenarios.

6. Task 4.2 – Frequency Domain Analysis (FFT)

Following the time-domain analysis, selected measurements were transformed into the frequency domain using the Fast Fourier Transform (FFT). The FFT decomposes a time-domain signal into its constituent frequency components, revealing which frequencies carry the most energy. This is essential for understanding the structural

dynamics (resonance frequencies) of the crash beam and the frequency characteristics of the hammer excitation.

The FFT plots (Figures 5–8) use a logarithmic frequency axis (10^0 to 10^5 Hz), allowing observation across a wide frequency range from low-frequency structural modes to the high-frequency range relevant to CISS detection (up to the accelerometer's rated range of 25 kHz).

6.1 Figure 5 – FFT of Task 3.2, Hit at Middle of Beam (~30 N)

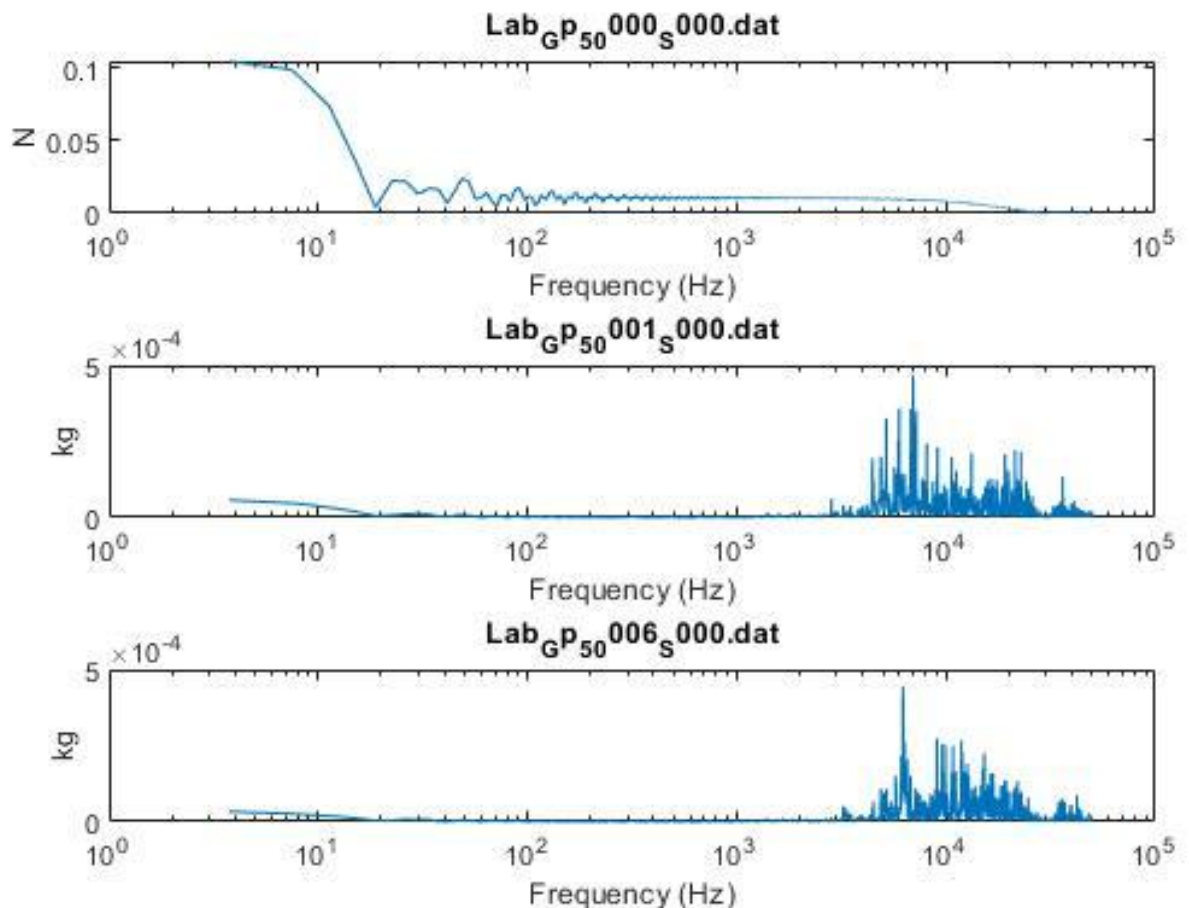


Figure 5: FFT frequency-domain signals — Task 3.2, Hit at middle of beam (light force). Top: Modal hammer [N vs. Hz]; Middle: Sensor 1 S001 [kg vs. Hz]; Bottom: Sensor 2 S006 [kg vs. Hz].

Figure 5 shows the FFT of the measurement from Figure 1 (light hit at the center, ~30 N). The modal hammer FFT (top panel) displays a characteristic broadband spectrum that is highest at low frequencies (near 1 Hz and below 100 Hz) and falls off gradually toward higher frequencies. This is typical of a modal hammer impact: the duration of the force pulse determines the frequency bandwidth excited. A shorter, harder pulse excites energy up to higher frequencies, while a softer, longer pulse rolls off sooner. Here, the hammer energy is concentrated below approximately 100–200 Hz.

The two sensor FFT spectra (middle and bottom panels) are more informative. Both S001 and S006 show relatively flat, low-level content in the low and mid-frequency

range (1 Hz to approximately 3 kHz), and then a pronounced cluster of spectral peaks between approximately 5 kHz and 30 kHz. These high-frequency peaks represent the characteristic resonant frequencies of the crash beam structure — the natural frequencies at which the beam preferentially vibrates when excited. The amplitude scale is $\times 10^{-4}$ kg, consistent with the relatively small input force (~ 30 N) of this measurement.

The similarity of the spectral envelopes between S001 and S006 confirms that both sensors are responding to the same structural modes of the beam. Minor differences in relative peak heights between the two sensors reflect differences in modal sensitivity at each sensor's mounting location — certain modes may have higher participation at one point than another.

6.2 Figure 6 – FFT of Task 3.2, Hit at Side of Beam (~ 150 N)

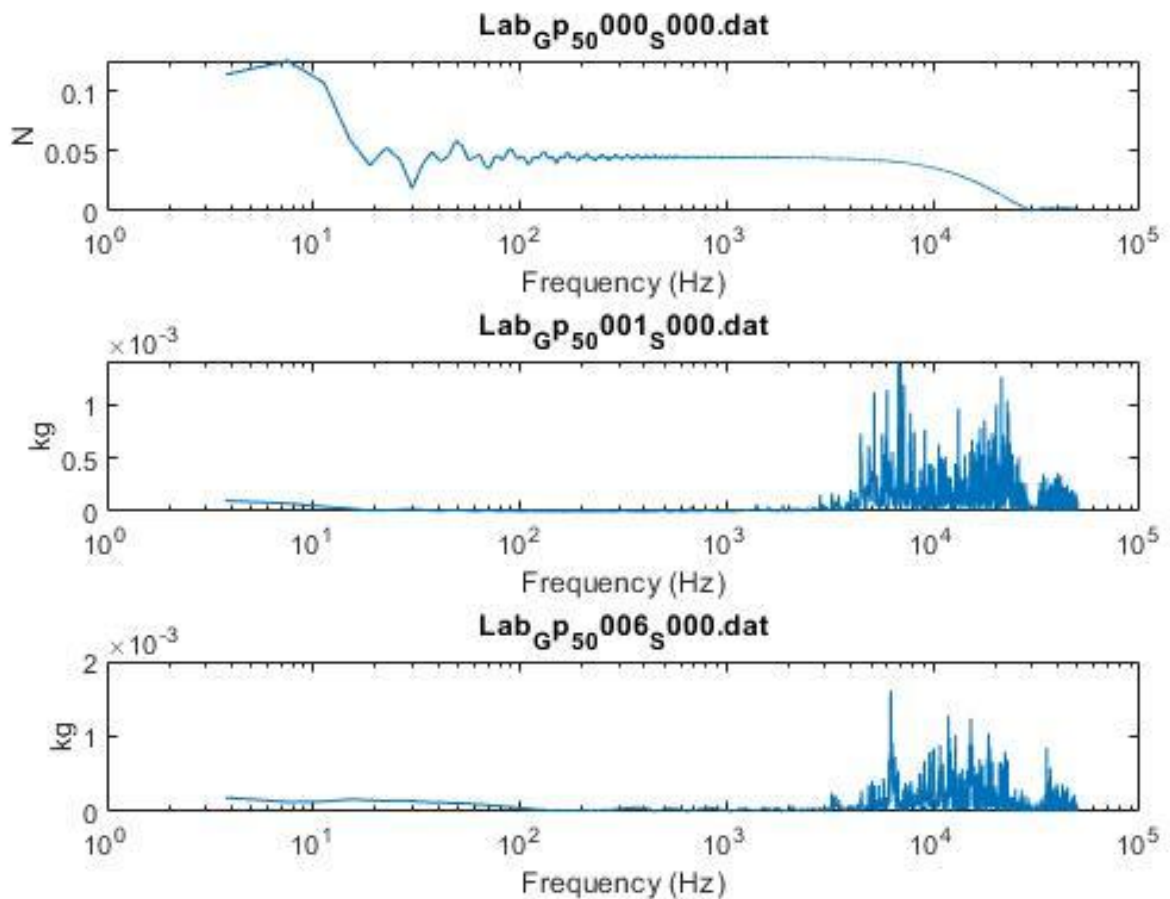


Figure 6: FFT frequency-domain signals — Task 3.2, Hit at side of beam (stronger force). Top: Modal hammer [N vs. Hz]; Middle: Sensor 1 S001 [kg vs. Hz]; Bottom: Sensor 2 S006 [kg vs. Hz].

Figure 6 shows the FFT of the side-beam hit measurement (Figure 2, ~ 150 N). The hammer FFT (top panel) again shows a broadband spectrum, but now with higher overall amplitude due to the greater input force. The hammer spectrum maintains energy up to approximately 200 Hz before rolling off, consistent with the duration of the impact pulse.

The sensor FFT spectra (both panels, amplitude scale $\times 10^{-3}$ kg) show significantly higher amplitudes than in Figure 5 — approximately 10 times larger — consistent with the $5\times$ increase in impact force and the generally nonlinear relationship between force and measured structural response magnitude (the greater force may also excite higher-order modes more effectively). The characteristic resonant frequency cluster between 5 kHz and 30 kHz is again prominent in both sensors.

Importantly, the positions of the resonant frequency peaks are consistent with those seen in Figure 5. This confirms that the resonant frequencies are a property of the beam structure itself (mass and stiffness distribution), not of the hit position or force level. What changes between different hit positions is the relative excitation of different modes — some modes are better excited from the center, others from the sides — which is reflected in subtle differences in relative peak amplitudes.

6.3 Figure 7 – FFT of Task 3.3, Light Hit at Middle (~20 N)

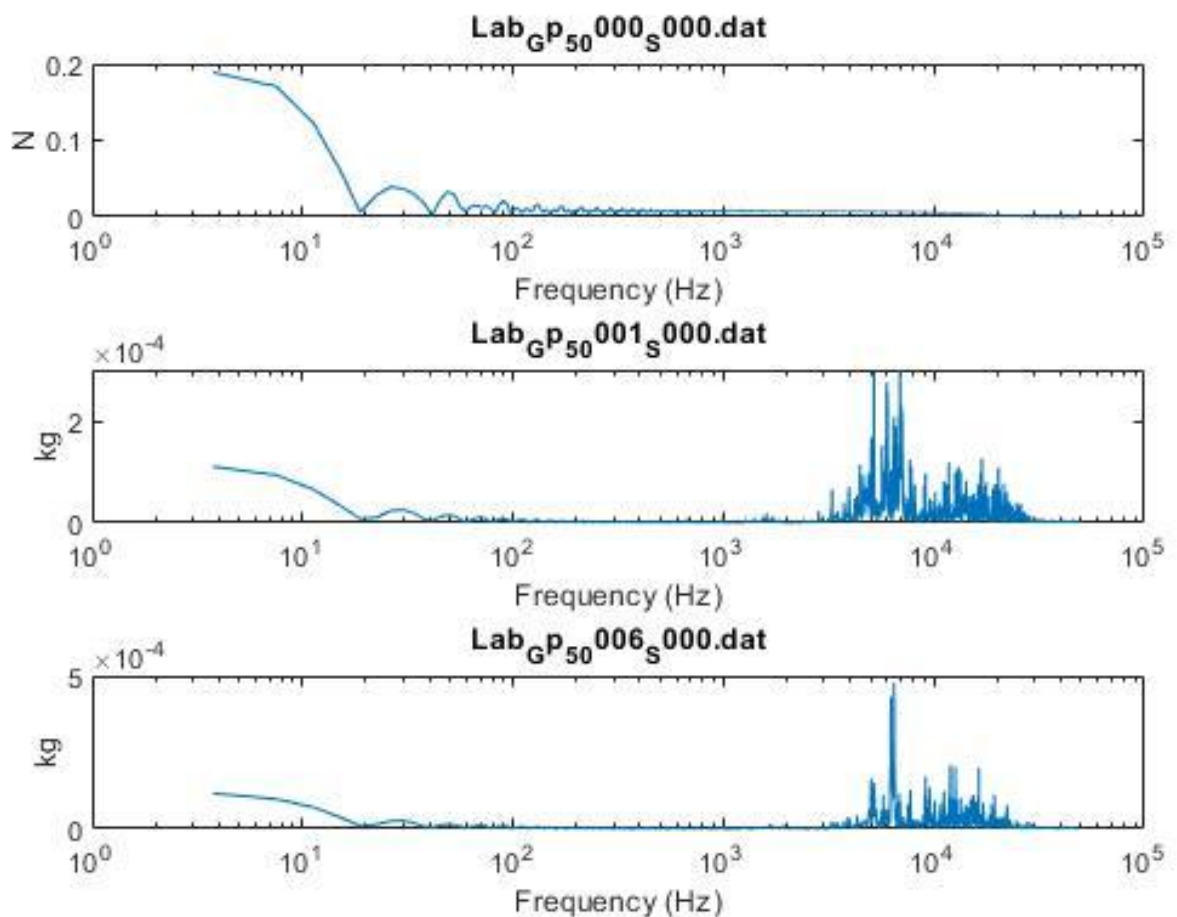


Figure 7: FFT frequency-domain signals — Task 3.3, Light hit at middle of beam. Top: Modal hammer [N vs. Hz]; Middle: Sensor 1 S001 [kg vs. Hz]; Bottom: Sensor 2 S006 [kg vs. Hz].

Figure 7 shows the FFT corresponding to the lightest hammer strike in Task 3.3 (~20 N at the center). The hammer spectrum (top panel) shows the lowest overall amplitude of all FFT measurements, consistent with the smallest input force. The spectrum extends to approximately 100–200 Hz before significantly attenuating.

The sensor spectra (scale $\times 10^{-4}$ kg) again clearly show the resonant peak cluster in the 5–30 kHz range. Sensor 2 (S006, bottom panel) shows somewhat higher spectral amplitudes at the resonant frequencies than Sensor 1 (S001), which is consistent with the time-domain observation in Figure 3 where S006 also showed slightly higher peak amplitudes. This further suggests that S006's mounting location is mechanically advantageous for detecting vibrations excited at the beam midpoint.

The FFT results confirm that even with a light 20 N impact, the resonant modes of the beam are clearly excited and detectable above the noise floor. This has practical CISS implications: real crash events of even moderate severity will excite these structural resonances strongly, making them reliably detectable.

6.4 Figure 8 – FFT of Task 3.3, Strong Hit at Middle (~100 N)

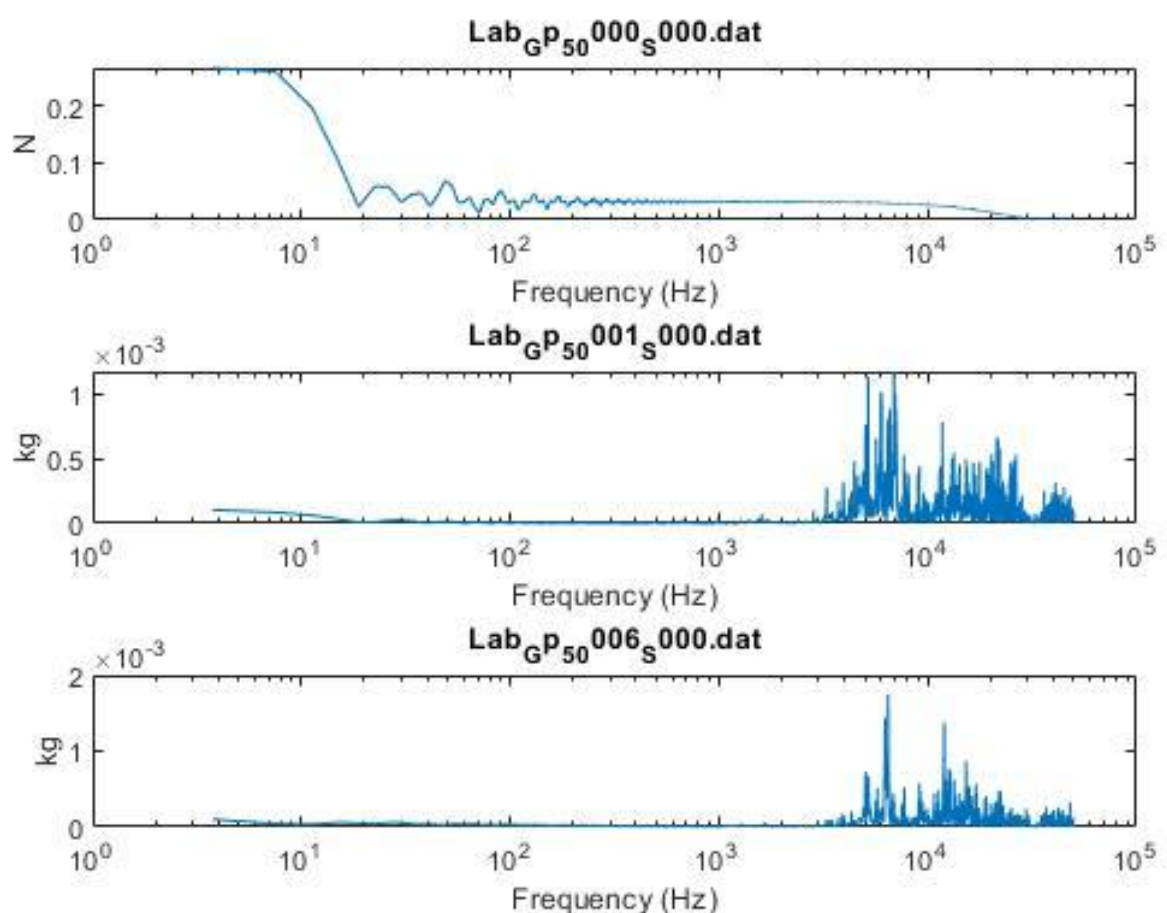


Figure 8: FFT frequency-domain signals — Task 3.3, Strong hit at middle of beam. Top: Modal hammer [N vs. Hz]; Middle: Sensor 1 S001 [kg vs. Hz]; Bottom: Sensor 2 S006 [kg vs. Hz].

Figure 8 represents the strongest center hit (~100 N) and shows the highest overall spectral amplitudes among the Task 3.3 measurements. The hammer FFT (top panel) shows a broadband spectrum reaching a peak amplitude of approximately 0.3 N and extending to beyond 100 Hz. Both sensor spectra (scale $\times 10^{-3}$ kg) display the now-familiar resonant peak cluster above 5 kHz, with peak amplitudes approximately 5–10 times larger than in Figure 7.

The proportional scaling of the FFT amplitudes between Figure 7 (~20 N, scale $\times 10^{-4}$) and Figure 8 (~100 N, scale $\times 10^{-3}$) further confirms the linear behavior of the system: multiplying the input force by ~5 multiplies the spectral output by approximately the same factor. The resonant frequency positions remain identical, as expected.

Comparing Figures 7 and 8, it is notable that the cluster of dominant resonant peaks in the 5–30 kHz range appears at the same frequencies in both measurements. This stability of resonant frequencies across different force levels is the physical basis for CISS pattern recognition: the resonant signature of a real crash would excite these same structural modes, and the system can be trained to recognize this characteristic frequency fingerprint as a crash event.

7. Evaluation Questions – Frequency Domain

7.1 Discussion of FFT Signals

The FFT signals across all four measurements (Figures 5–8) are correctly and meaningfully converted. The following key observations confirm this:

- The modal hammer FFT correctly shows a broadband (wideband) frequency content characteristic of an impulsive force input. The energy falls off gradually from low frequencies toward high frequencies, with the rolloff frequency approximately related to the pulse duration (shorter pulse = higher bandwidth).
- Both accelerometer sensors consistently reveal the same structural resonant frequency cluster in the range of approximately 5–30 kHz. These peaks appear at the same frequencies across all four measurements regardless of hit position or force level, confirming they are true resonant modes of the crash beam structure rather than measurement artifacts.
- The amplitude scaling of the FFT spectra is consistent with the input force levels — larger input produces proportionally larger output, confirming linear system behavior and correct sensor calibration.
- No unexpected spectral artifacts, aliasing, or unphysical content (e.g., spurious peaks at 50 Hz powerline frequency) are visible in the sensor spectra, indicating clean signal conditioning and proper grounding in the measurement setup.

7.2 Frequency Created by the Hammer

The modal hammer does not generate a single discrete frequency but rather a broadband frequency content, as can be seen from the smoothly decaying FFT in the top panel of each FFT figure. The energy is concentrated at low frequencies (below approximately 100–200 Hz), which corresponds to the quasi-static nature of the impact force pulse. The useful frequency bandwidth of the hammer excitation in this experiment is approximately 1 Hz to 200 Hz. This means the hammer reliably excites all structural modes of the beam that fall below this frequency range, while higher-frequency resonances (like those observed at 5–30 kHz in the sensor spectra) are excited by the structural wave propagation rather than directly by the hammer's force spectrum.

7.3 Characteristic Resonant Frequencies of the CISS Sensors

Both CISS accelerometers (Sensor 1 S001 and Sensor 2 S006) exhibit a prominent cluster of resonant peaks in the frequency range of approximately 5,000 Hz to 30,000 Hz (5 kHz to 30 kHz). Within this cluster, dominant peaks are visible at approximately:

- ~7,000–10,000 Hz: A broad group of lower-amplitude peaks, representing lower-order bending and torsional modes of the beam
- ~15,000–20,000 Hz: The most prominent resonant peak cluster, likely corresponding to higher-order flexural modes and longitudinal modes of the crash beam
- ~25,000–30,000 Hz: Highest-frequency peaks near the upper limit of the accelerometer's specified frequency range (10–25 kHz), potentially including both structural resonances and the sensor's own mechanical resonance

These frequencies are characteristic of the specific crash beam geometry, wall thickness, and material. In a real CISS implementation, the ECU is programmed to monitor energy in precisely these frequency bands as part of the crash detection algorithm.

8. General Questions

8.1 Frequency Resolution in FFT — Two Main Factors

The resolution in the frequency domain (i.e., the smallest difference between two frequencies that can be distinguished in the FFT output) depends primarily on two factors:

- Signal duration / Record length (T): The frequency resolution Δf is inversely proportional to the total length of the time signal: $\Delta f = 1/T$. A longer measurement window produces a finer (better) frequency resolution. For example, a 1-second recording gives 1 Hz resolution, while a 0.1-second recording gives only 10 Hz resolution. This is why long recordings are preferable when distinguishing closely spaced resonant frequencies.
- Sampling frequency (Fs): The sampling rate determines the maximum frequency that can be represented in the FFT (the Nyquist frequency = $F_s/2$). A higher sampling rate extends the frequency axis further, allowing high-frequency content to be captured. However, it does not improve the frequency resolution per se — it only expands the range. The resolution remains governed by the record length. Together, F_s and T determine the number of FFT frequency bins ($N = F_s \times T$) and the spacing between them ($\Delta f = F_s/N = 1/T$).

8.2 Windowing in Frequency Analysis

Windowing is a signal processing technique in which the time-domain signal is multiplied by a smoothly tapering function (the 'window') before applying the FFT. The purpose of windowing is to reduce spectral leakage — a phenomenon that occurs when the signal to be analyzed does not complete an integer number of cycles within the recorded time window.

When an FFT is computed, it implicitly assumes that the recorded signal repeats periodically. If the signal does not start and end at the same value (i.e., there is a

discontinuity at the edges of the recording), the FFT interprets this discontinuity as a high-frequency artifact and spreads energy from a narrow spectral peak across adjacent frequency bins — this is spectral leakage. Applying a window function (such as the Hann, Hamming, or Flat-Top window) forces the signal to taper smoothly to zero at both ends, eliminating the edge discontinuity.

Windowing is necessary when: (1) the signal is continuous (non-impulsive) and does not naturally decay to zero within the measurement window, such as a steady sinusoidal vibration; or (2) when high spectral resolution and clean peak separation are critical. For impulsive signals — like those recorded in this CISS experiment — windowing is generally less critical, because the structural ring-down naturally decays to near-zero before the end of the recording window. However, applying a window function (e.g., Hann window) is still considered good practice to further reduce any residual leakage and improve the clarity of the resonant peak identification.

9. Conclusion

This experiment successfully demonstrated the fundamental principles of Crash Impact Sound Sensing (CISS) using a crash beam, a modal hammer, and two Kistler accelerometers connected to an LTT24 data acquisition system. The key findings are:

- Structure-borne sound propagates from the hammer impact point to both sensor locations within milliseconds, and the vibration ring-down lasts approximately 80 ms — well within the critical airbag deployment time window.
- Changing the hit position (Task 3.2) affects the relative amplitude and distribution of sensor responses, reflecting the spatial characteristics of structural wave propagation.
- Changing the hit force (Task 3.3) scales the sensor amplitude proportionally, confirming the linear behavior of the crash beam and sensor system within the tested range.
- FFT analysis reveals a characteristic resonant frequency signature of the crash beam in the 5–30 kHz range, which is independent of hit position and force level. This structural fingerprint is the basis for CISS crash detection algorithms.
- The propagation speed of structure-borne sound through metal is approximately 5,000 m/s, enabling CISS to detect crash events orders of magnitude faster than conventional methods — a critical advantage for timely airbag deployment and occupant safety.

The experiment validates the physical operating principle of CISS and illustrates why this sensing technology has become an important complement to traditional acceleration-based crash sensing in modern vehicles.